PART One

Preliminaries

CHAPTER 1

Introduction

This book deals with the fixed income markets and the best point to start is to define what this means. The book will follow market usage by defining fixed income instruments as contracts that specify payment obligations that are not linked to the economic situation of the obligor. It also excludes from the scope contracts that establish payments depending on the performance of physical assets, such as commodities or weather. The book will further discuss a range of fixed income derivatives, namely contracts that specify obligations that depend on the performance of fixed income instruments.

Fixed income instruments can be broadly divided in to bilateral contracts and securities. Securities differ from bilateral contracts in that they are transferrable without the consent of all parties. Note that some features of securities may be linked to bilateral contracts. Table 1.1 lists various fixed income instruments in these two categories.

In German, fixed income instruments are referred to as ‘Renten’ and the same word is also used to refer to pensions. In the English vernacular, the scope of the word ‘rent’ to signify income streams has been narrowed down significantly to refer to payments linked to the use of physical assets. The Latin translation of the word ‘rent’, ‘pension’, is now used in English to designate the income stream that accrues to people after a lifetime of work. On the other hand, ‘Pension’ in financial German refers to the renting out of financial assets[1]. This etymological divergence between otherwise related languages highlights an important point, namely that the notion of fixed income instruments is not new but underwent several changes of connotation over time.

TABLE 1.1 Non-exclusive list of instrument types

Bilateral contracts	**Securities**
Deposits	Bonds
Loans	Commercial paper
Swaps	Certificates of deposit
Futures	Asset-backed securities

[1]It has to be admitted that few people today use the word Wertpapierpensionsgeschäft.

Apart from commodity contracts, the definition above rules out common and preferred equity instruments where the issuer has some discretion about the timing and amount of payments. This is a strong restriction because the party to the contract that is obliged to make a payment has no rights versus the payee. The very word ‘equity’ implies an equitable relationship between payer and payee, whereas in the fixed income setting, the obligor usually has limited discretion.

The ancient Greeks despised money lending, as evidenced in Aristotle’s Politics:

> There are two sorts of wealth-getting, as I have said; one is part of household management, the other is retail trade: the former necessary and honourable, while that which consists in exchange is justly censured; for it is unnatural, and a mode by which men gain from one another. The most hated sort, and with the greatest reason, is usury, which makes a gain out of money itself, and not from the natural object of it. For money was intended to be used in exchange, but not to increase at interest.

The three book religions, Judaism, Christianity, and Islam, severely restrict the lending of money against interest and hence the majority of fixed income instruments. In scriptural terms, only Judaism permits charging interest at all, and only in the case of Jews charging interest to gentiles according to the usual intepretation of Deuteronomy 23:19–20:

> Thou shalt not lend upon usury to thy brother; usury of money, usury of victuals, usury of anything that is lent upon usury. Upon a stranger thou mayest lend upon usury; but unto thy brother thou shalt not lend upon usury; that the LORD thy God may bless thee in all thou settest thine hand to in the land whither thou goest to possess it.

For Christians, the encyclical Vix pervenit [88] issued by Benedict XIV. on 1 November 1745 opened a legal distinction between usury and lending money at interest, although it reaffirmed the general prohibition of usury. Even earlier, the Pax Westphalica [67] in 1648 casually refers to fixed interest obligations. Today, the issue presents no particular problem for Christians. However, many countries retain prohibitions on excessive interest charges that relate back to usury concepts.

In contrast, Islamic law (*Shar’iah*) has retained its prohibition to lending at interest (*riba*), or indeed unconditional repayment obligations. Together with *gharar* (uncertainty) and *maysir* (gambling), such practices are strictly forbidden by the Qur’an. This rules out investing money for interest (but not the earning of profits from equity investments) and also the concept of pure options. These restrictions present an obstacle for Muslim investors to interact with the western financial system. Accordingly, a system of Islamic finance has developed that allows profitable investments at low risk.

In order to understand why fixed income is apparently such a new concept, it is instructive to study the prohibitions above, setting aside the issue of divine inspiration. For instance, in Exodus 22:25, Jews are instructed:

> If thou lend money to any of my people that is poor by thee, thou shalt not be to him as an usurer, neither shalt thou lay upon him usury.

The implication is that a lender of money is somebody with surplus cash while a borrower is in an emergency situation and needs cash to make ends meet. Exploiting such an emergency situation is morally reprehensible and therefore forbidden. Aristotle's aversion to lending is similarly explained in that he abhors the idea of a necessary social process being abused for pecunary gain. Human beings have to trade in order to exist, but the act of trading does, in his eyes, not contribute to the production of goods. Profiting from something that is not actually increasing the store of wealth of humankind is therefore despicable.

These arguments do not sound too unfamiliar today and debts are still viewed negatively in some parts of the political spectrum [43]. The innovation of some newer critical authors is to draw a distinction between social convention of debts that have to be served unconditionally on one hand, and more flexible moral notions of obligations on the other. The latter are alleged by these authors to be more in line with human development. Setting aside the question whether there is indeed an anthropological basis for distinguishing between morals and social convention, there remains a question as to whether lenders are willingly absorbing an ambiguity in their relationship with borrowers, or instead view their claims as legally absolute. An argument made in Henry V [74] (4.1) may be instructive:

> King Harry So, if a son is by his father sent about merchandise do sinfully miscarry upon the sea, the imputation of his wickedness, by your rule, should be imposed on his father, that sent him. Or if a servant, under his master's command transporting a sum of money, be assailed by robbers, and die in many irreconciled iniquities, you may call the master the author of the servant's damnation. But this is not so. The King is not bound to answer the particular endings of his soldiers, the father of his son, nor the master of his servant, for they purpose not their deaths when they propose their services.

Similarly, most lenders do not 'purpose' inability to repay when they lend money and indeed tend to prefer to lend to borrowers who repay on schedule. A pensioner who has invested in a mortgage does not seek the destitution that may befall the borrower who is dispossessed upon default and would usually prefer not to face the risk of his or her own destitution should the process of liquidating the property faily to reproduce the originally expected income stream.

What has changed in recent times is that lenders are no longer necessarily rich and borrowers are not necessarily poor. People now generally accept that individuals need to put money aside to prepare for their old age and that if they were unable to do so without being paid interest, inflation and credit risk would erode the value of their savings. They also have no reason to enter a relationship based on equity with whoever has use of their money at the time. The association of fixed income with pension in the German word 'Rente' is therefore now quite natural. On the borrower side, it is now also common to borrow money for convenience, such as in a mortgage, or to invest in a new business. Many business owners do not want to form equitable partnership with capital providers. They prefer to pay interest on that capital and retain the profits they expect from their venture. In addition, interest payments are tax-deductible in many jurisdictions. In the times when the great religious books were written, large multigenerational families would have handled equivalent transfers internally, whereas businesses nowadays have to conclude external contracts.

The sometimes shady area of consumer finance shows, however, that inequity is still an issue in money lending. This means that the debate about the moral implications of fixed income liabilities is not likely to end. So-called predatory lenders, who do seek out borrowers that can be expected not to repay and can therefore be pursued for their assets following default, do exist. Most modern legal codes provide some safeguards against such abuses of the legal system, such as upper limits on interest rates, or know-your-client rules.

At the same time, some supposedly new arguments about wealth distribution can be traced to insufficient awareness of what are essentially fixed income assets. Popular studies of wealth disparity tend to focus on traceable forms of assets, such as accumulated income, real estate, etc [68]. However, Mr Darcy is described in 'Pride and Prejudice' as having £10,000 a year. Earlier, Sir Toby in 'Twelfth Night' justifies the description of Sir Andrew Aguecheek as 'as tall a man as any in Illyria' with 'he has three thousand ducats a year'. This description of a person's assets in terms of the annuity they generate, instead of their current value, was quite common at the time these books were written. Using this approach today can be eye-opening.

Mayer Amschel Rothschild, founder of the Rothschild banking dynasty and arguably one of the richest men in the world at his time, died on 16 September 1812 of an infection [35]. An affliction of this kind would be trivial to cure today, and that cure would be free of charge to even the poorest British citizen. The National Health Service (NHS) of the United Kingdom had an annual budget of £116.4bn in 2015/16, equivalent to around £1,800 per head of the UK population. The NHS is free of charge at the point of use, so every person registered with the NHS has the right to consume services, if they require, that are worth around £1,800 per year. Assuming for illustration an average remaining life span of 30 years and a discount rate of 1.8% (the long-term yield at that time), a person wishing to obtain an equivalent income stream would have to invest an amount of about £41,400 using the annuity formula Equation (16.1). Simply by registering with the NHS, an average UK resident therefore in effect acquires an asset that is equivalent to £41,400. This asset is of course only valuable to the extent that illness creates a need to draw on it. The point is, though, that a person *outside* the NHS would have to accumulate that amount of wealth to be able to purchase the services and goods available to the average NHS user.

Although the NHS-implied assets are invisible on citizens' balance sheets, their counterpart on the national balance sheet of the United Kingdom is not. The NHS has to be funded either through taxes or government debt. Of the latter, the United Kingdom carried £29,400 per head of the population in 2016.

Germany provides a monthly stipend that in 2017 was €409 and occasional additional payments to single people with no dependants unconditionally on previous earned income. Using current 30 year interest rates, a 30 year annuity of this magnitude had a present value of €127,000. A person with savings of this amount would be considered reasonably well off even though few would consider the equivalent income stream as attractive. Absent this welfare system, one would have to accumulate this very substantial sum to obtain an equivalent income stream. Universal free or subsidised education is another example of a valuable claim on society that is in principle open to all but not valued as an asset.

Classically, wealthy citizens were called upon to fund common expenses and in return earned rights to future distributions of national wealth [57], for instance through the interest on government bonds. Although wealthy persons have been encouraged to provide for the poor by way of charity in many ethical frameworks, this has always been an exhortation rather than an obligation. From the viewpoint of the recipients, such charitable giving was a blessing, not a right. Traditionally, therefore, societies put a lot of value on either income-earning assets or opportunities for work that would create meaningful income streams. The onus to pay for goods and services, in other words, has traditionally been on the consumer.

Modern societies legislate unconditional claims on society that are not the result of prior contributions, such as welfare or universal health care. Because such liabilities of the national budget ultimatively have to be backed by corresponding tax assets, tax payers, who are the ultimate contingent debtors in this relationship, may at some point feel aggrieved. Claimants on social redistribution in effect have claims on taxpayers and government bond investors that resemble those that creditors have on debtors, namely claims on a fixed income stream.

No extreme point in this virtual creditor–debtor relationship presents an obvious optimum for overall welfare. Denying any form of redistribution would not only create social unrest, but would in most ethical frameworks be considered immoral. It would also be economically ruinous because it would consign potential consumers to destitution, reducing the scope for producers to earn an income. Taken too far, redistribution on the other hand would instead discourage producers from earning (or at least declaring) taxable income. The optimal extent of redistribution is open to debate (known in supply-side economics as the Laffer curve), but does most certainly depend not only on general measures such as welfare expenditure relative to gross domestic product, but also on the specific form in which redistribution takes place [27].

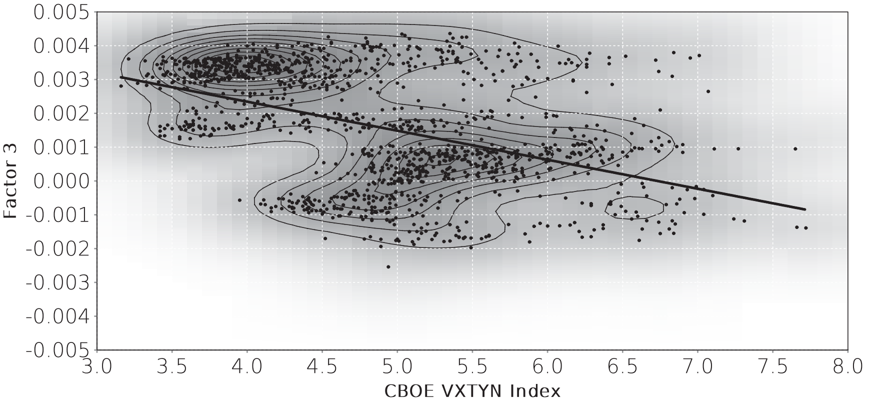
Islamic finance, which is developing borrowing and lending instruments based on *shar'iah* notions of equity, could provide interesting lessons in this regard. The pricing of Islamic banking products can generally be translated into terms that would be familiar to Western bankers, but the legal relationships between lenders and borrowers follow different rules. As Muslim nations grow in economic importance and the Islamic finance sector expands, at some point it will perhaps start to influence Western notions of banking. Already at this time, the Ecological and Social Responsibility (ESR) investment trend shares some similar ideas.

This book has been designed as course material and, with the exception of the last chapter, can be read from front to back. Industry practitioners might hopefully find it useful as a general vade mecum afterwards. The chapter arrangement follows a fairly conventional structure. Beginning with money, banking and monetary policy (Chapters 2–9) it then moves to how transactions take place (Chapters 10–12). Having thus covered the infrastructure of financial markets, the next part deals with the money and bond markets (Chapters 13–18). Several ideas linked to valuation follow in Chapters 19–22. The next parts, presented in Chapters 23–27 are detours into more complex parts of the bond market, dealing with inflation, credit risk, and mortgage prepayments. The essential derivatives markets, bond futures and swaps are presented in Chapters 28 and 29. With the complete set of instruments in hand, Chapters 30–32 introduce trading strategies, while Chapters 33–38 address standard portfolio and risk management strategies. Chapter 39 is a reference to various bond markets provided mostly to illustrate the variety of products that exist.

As a general approach, this book does not only present and explain concepts but also critically discusses their shortcomings. Like any human endeavour, fixed income analysis uses imperfect tools. Imperfections do not invalidate tools, but make it incumbent on their user to know when not to trust them.

A quick word on charts

This book uses a particular form of scatter plot, for instance the one shown here.



In this plot, the actual observations are augmented by information about their probability density. When many points fall closely together in the same area, the human eye tends to see them as just one point. This optically overvalues outliers. In the scatter plots here, a two-dimensional Gaussian is calculated for each individual data points using the observed variance in the x and y direction of the chart for all data points. These individual Gaussians are then evaluated on a regular grid and added up for all data points. The resulting probability density function is represented in a greyscale behind the usual dots of the scatter plot. In line with topographic maps, isolines are added to give additional flavour to the resulting shape.